



Medical imaging chip

Researchers have developed a powerful pint-size millimeter-wave signal generator. <http://digit.in/pintChip>



Nintendo Switch stolen

Nintendo claims that a Switch device that was used in a hands-on video had been stolen from a distributor. <http://digit.in/Switchsteal>

104 NOT OUT! ISRO'S PSLV-C37 MISSION

The complete lowdown of a historic Indian space milestone that everyone is talking about

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On February 15, 2017, India made its entry into the books of yet another world record in space missions. The Indian Space Research Organisation (ISRO) executed a successful PSLV C-37 launch mission, involving the deployment of a staggering 104 satellites into space. Starting off its journey at India's spaceport in Sriharikota, Andhra Pradesh, the payload included three Indian satellites, along with 96 American, and a satellite each from Israeli, Kazakh, Swiss, Emirati & Dutch organisations. The entire payload weighed around 1,378 kilograms. The previous record of 37 satellites by the Russian Dnepr rocket set back in June 19, 2014 was decimated in massive proportions. The PSLV-34 also left behind its own previous record set in 2016, by more than five times, standing with a payload of 20 satellites. The success of the mission attracted praises from across the globe. Revered spaceman and SpaceX founder, Elon Musk, also joined in to acknowledge the "impressive" achievement. It was a wonderful sight to witness the team of scientists momentarily leaving behind their otherwise stern and focused state to a group of overjoyed kids celebrating their pioneering success.



The PSLV-C37 sits on the First Launch Pad at the Satish Dhawan Space Centre - calm before the Storm

PSLV

Launching 104 satellites to space is no joke and there's no extra points in guessing what took over this responsibility from ISRO's arsenal of launch systems. As expected, the venerable Polar Satellite Launch Vehicle (PSLV) was chosen for this mission. Already basking in the glory of numerous successful missions, the recent PSLV launch makes it a total of 37. India was facing a serious communication gap during 2011 with demands growing as satellites neared towards the end of their operational life. The unavailability of spare transponder capacity pushed the launch of the GSAT-12 satellite. One motivating factor behind this launch was the inability of the European Arianespace SA which is

usually the launch provider for communication satellites. This resulted into some rejigging of the PSLV for a "fast track" mission, launching to the higher Geostationary Transfer Orbit (GTO). Consequently, on July 15, 2011, the PSLV-C17 mission took off with the GSAT-12, delivering yet another flawless performance, mitigating the crisis. Encouraged by the PSLV's performance and success rate, satellite makers from 23 countries have bestowed their faith on ISRO so far.

The PSLV programme was conceived in the 1980s towards the country's first attempt to develop meaningful Space Transportation System (STS). The 70's and 80's witnessed the development of the Satellite Launch Vehicle 3 (SLV 3) and the Augmented Satellite Launch Vehicle (ASLV), eventually building up a repository of institutional knowledge in rocket and space technology. Years of experience helped setup the physical infrastructure required that made the latest achievement possible.

Reaching the polar Sun Synchronous Orbit (SSO) – 600-1000 km over the Earth surface – was the prime objective of ISRO's PSLV project to launch its Indian Remote Sensing (IRS) satellites. This is a very crucial orbit, in consonance with ISRO's mandate to harness space for socio-economic development. With the PSLV in its arsenal, India possesses the capability of launching its own remote sensing "observation" satellites. The wide range of its functions include weather monitoring, tracking forest cover, determining fish density & discovering deposits of natural resources.

Data from ISRO's Oceansat-2 satellite enabled meteorologists to gauge super typhoon Haiyan's severity & path,